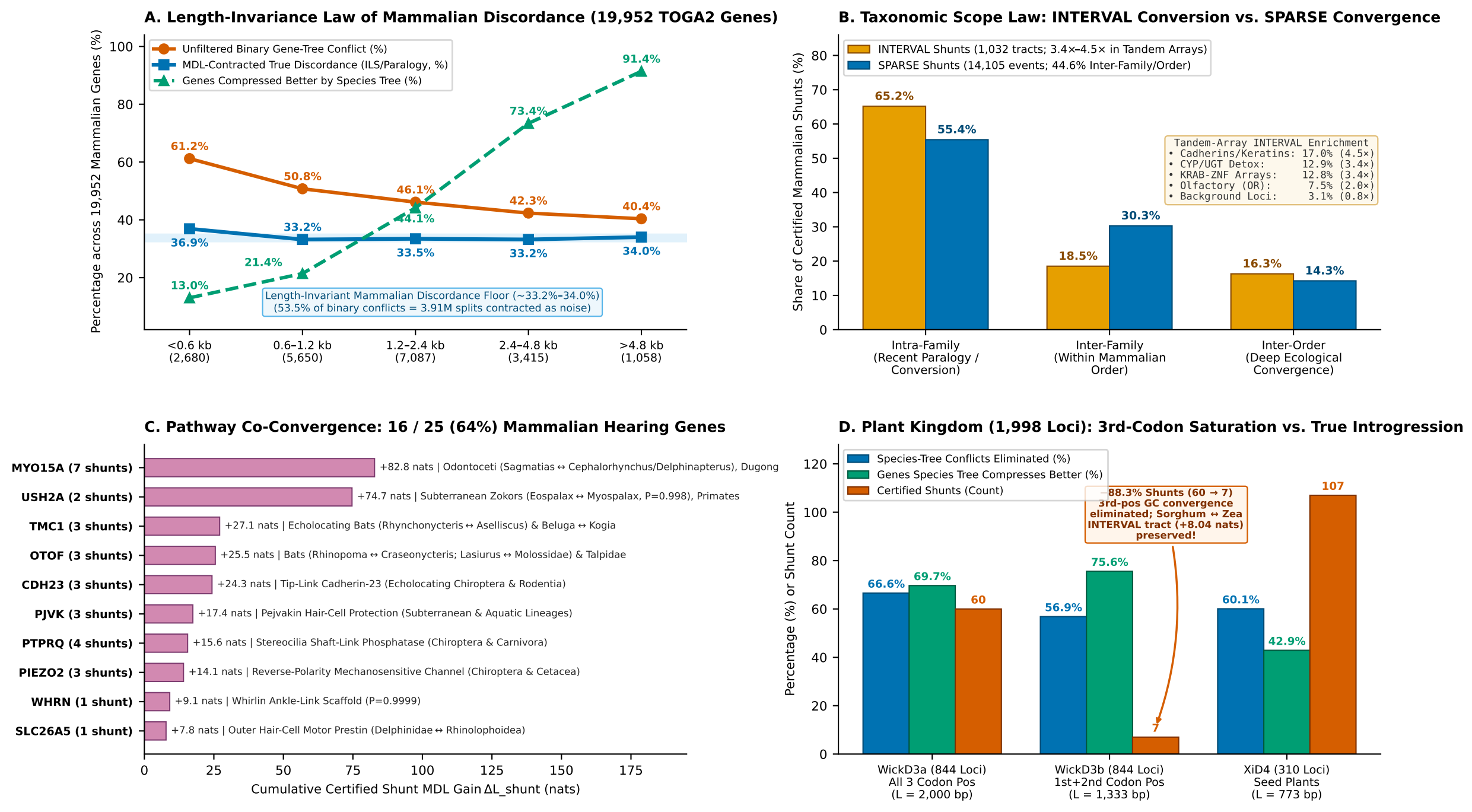

A. Length-Invariance Law of Mammalian Discordance (19,952 TOGA2 Genes)
Unfiltered Binary Gene-Tree Conflict (%)
MDL-Contracted True Discordance (ILS/Paralogy, %)
Genes Compressed Better by Species Tree (%)
Percentage across 19,952 Mammalian Genes (%)
61.2%
50.8%
46.1%
42.3%
40.4%
36.9%
33.2%
33.5%
33.2%
34.0%
13.0%
21.4%
44.1%
73.4%
91.4%
Length-Invariant Mammalian Discordance Floor (~33.2%–34.0%)
(53.5% of binary conflicts = 3.91M splits contracted as noise)
<0.6 kb (2,680)
0.6–1.2 kb (5,650)
1.2–2.4 kb (7,087)
2.4–4.8 kb (3,415)
>4.8 kb (1,058)
B. Taxonomic Scope Law: INTERVAL Conversion vs. SPARSE Convergence
INTERVAL Shunts (1,032 tracts; 3.4×–4.5× in Tandem Arrays)
SPARSE Shunts (14,105 events; 44.6% Inter-Family/Order)
Share of Certified Mammalian Shunts (%)
65.2%
55.4%
18.5%
30.3%
16.3%
14.3%
Tandem-Array INTERVAL Enrichment
• Cadherins/Keratins: 17.0% (4.5×)
• CYP/UGT Detox: 12.9% (3.4×)
• KRAB-ZNF Arrays: 12.8% (3.4×)
• Olfactory (OR): 7.5% (2.0×)
• Background Loci: 3.1% (0.8×)
Intra-Family (Recent Paralogy / Conversion)
Inter-Family (Within Mammalian Order)
Inter-Order (Deep Ecological Convergence)
C. Pathway Co-Convergence: 16 / 25 (64%) Mammalian Hearing Genes
MYO15A (7 shunts)
+82.8 nats | Odontoceti (Sagmatias ↔ Cephalorhynchus/Delphinapterus), Dugong
USH2A (2 shunts)
+74.7 nats | Subterranean Zokors (Eospalax ↔ Myospalax, P=0.998), Primates
TMC1 (3 shunts)
+27.1 nats | Echolocating Bats (Rhynchonycteris ↔ Aselliscus) & Beluga ↔ Kogia
OTOF (3 shunts)
+25.5 nats | Bats (Rhinopoma ↔ Craseonycteris; Lasiurus ↔ Molossidae) & Talpidae
CDH23 (3 shunts)
+24.3 nats | Tip-Link Cadherin-23 (Echolocating Chiroptera & Rodentia)
PJVK (3 shunts)
+17.4 nats | Pejvakin Hair-Cell Protection (Subterranean & Aquatic Lineages)
PTPRQ (4 shunts)
+15.6 nats | Stereocilia Shaft-Link Phosphatase (Chiroptera & Carnivora)
PIEZO2 (3 shunts)
+14.1 nats | Reverse-Polarity Mechanosensitive Channel (Chiroptera & Cetacea)
WHRN (1 shunt)
+9.1 nats | Whirlin Ankle-Link Scaffold (P=0.9999)
SLC26A5 (1 shunt)
+7.8 nats | Outer Hair-Cell Motor Prestin (Delphinidae ↔ Rhinolophoidea)
Cumulative Certified Shunt MDL Gain ΔL_shunt (nats)
D. Plant Kingdom (1,998 Loci): 3rd-Codon Saturation vs. True Introgression
Species-Tree Conflicts Eliminated (%)
Genes Species Tree Compresses Better (%)
Certified Shunts (Count)
Percentage (%) or Shunt Count
66.6%
69.7%
60
56.9%
75.6%
7
60.1%
42.9%
107
−88.3% Shunts (60 → 7)
3rd-pos GC convergence eliminated; Sorghum ↔ Zea INTERVAL tract (+8.04 nats) preserved!
WickD3a (844 Loci) All 3 Codon Pos (L = 2,000 bp)
WickD3b (844 Loci) 1st+2nd Codon Pos (L = 1,333 bp)
XiD4 (310 Loci) Seed Plants (L = 773 bp)